Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Mechatronics Engineering

Course Title	:	Electronics Sessional
Course Code	:	EEE 2188
Experiment No	:	01
Name of Experiment	:	Familiarization of different Equipment involved with Electronics Sessional
Date of Experiment	:	21 June 2026
Date of Submission	:	28 June 2026

Submitted By	**Submitted To**
Name : TASNIMUL HASAN Roll : 2408020 Series : 24	Md. Firoj Ali Associate Professor Department of Mechatronics Engineering RUET Prangon Das Assistant Professor Department of Mechatronics Engineering RUET

Experiment No: 1

Name of Experiment: Familiarization of different Equipment involved with Electronics Sessional

1.1 Theory

The study of electronics begins with the visualization and measurement of electrical signals, primarily through the use of an oscilloscope [1]. The basic function of this instrument is to generate an X-Y plot that displays changes in voltage at its input relative to time [1]. The X-axis represents time, typically measured in milliseconds or microseconds, while the Y-axis represents the difference in electric potential, or voltage [1]. This allows for the continuous observation of various waveforms, such as sine waves and square waves, and the measurement of critical parameters like frequency, period, and peak-to-peak voltage [1]. To ensure a steady and readable display, a trigger level must be established, which sets the specific voltage point at which the oscilloscope starts collecting and displaying the signal [1].

Figure 1.1: Theoretical X-Y plot of an oscilloscope display showing a sinusoidal signal with labeled time and voltage axes.

Electronic circuits are composed of passive and active components that interact according to fundamental principles like Ohm's Law, which states that the voltage across a resistor is proportional to the current passing through it [2]. While resistors dissipate energy as heat, ideal capacitors and inductors are designed to store energy within electric and magnetic fields, respectively [3]. Active components, such as Bipolar Junction Transistors (BJTs) and Operational Amplifiers (Op-Amps), provide the capability to increase the strength of weak signals through amplification or to control the flow of current as switches [4], [5]. The Operational Amplifier, a high-gain DC differential amplifier, is particularly versatile because its performance and functions—such as integration, differentiation, or summation—are determined by the external feedback components connected between its output and input terminals [5].

Figure 1.2: Block diagram of a differential amplifier system illustrating the principle of amplification and feedback.

The successful operation of semiconductor devices depends on biasing, which is the application of specific DC voltages to establish the desired operating mode [4]. For instance, a BJT operates in the active mode for amplification when its emitter-base junction is forward-biased and its collector-base junction is reverse-biased [4]. Similarly, diodes are unidirectional devices that allow current to flow only when forward-biased, whereas Zener diodes are specifically engineered to operate in the reverse breakdown region to provide stable voltage regulation [6]. Understanding these operational modes is essential for laboratory work, as it enables the practitioner to develop an intuition for expected measurement values and to identify anomalies in circuit behavior [1], [4].

Figure 1.3: Idealized characteristic curve (I-V curve) showing the theoretical behavior of a semiconductor junction under different biasing conditions.

1.2 Objectives

1. To learn about the commonly used equipment in the Electronics Sessional.
2. To observe the various parameter of the electric circuit.
3. To observe the different instrument which are usually need for measurement.
4. To understand the use of the instrument and components in experiment.

1.3 Major Components and Equipment

1.3.1 Resistor

Definition

A resistor is a passive two-terminal electronic component that implements electrical resistance as a circuit element [2]. It obeys Ohm's Law ($V = IR$), meaning the voltage drop across it is directly proportional to the current flowing through it. Resistance is measured in ohms (Ω), and resistors values are most commonly indicated by a system of color-coded bands printed on their body.

Figure 1.4: Resistor and color-band coding chart

Specifications

The primary specifications of a resistor are its resistance value (ranging from fractions of an ohm up to several gigaohms), its tolerance (the permissible deviation from the nominal value, commonly $\pm1\%$, $\pm5\%$, or $\pm10\%$), and its power rating (the maximum heat it can safely dissipate, typically $\frac{1}{8}$ W to 2 W for through-hole types and up to several hundred watts for industrial power resistors) [7]. The temperature coefficient of resistance (TCR), expressed in parts per million per degree Celsius (ppm/°C), quantifies how much the resistance drifts with temperature; carbon-film resistors exhibit TCR values of ± 100 to ± 500 ppm/°C, while metal-film types achieve as low as ± 25 ppm/°C, and precision metal-foil resistors can reach ± 0.2 ppm/°C [7]. Resistors are available in through-hole (axial lead) and surface-mount (SMD) packages; SMD types are labeled using a three- or four-digit numeric code rather than color bands.

References

[1] B. Korst and M. Zaharia, *An introduction to oscilloscopes*, Technical manual, n.d.

[2] Wikipedia. "Resistor." Last edited: May 30, 2026. [Online]. Available: https://en.wikipedia.org/wiki/Resistor

[3] n.d., *Capacitor and inductors*, Unpublished technical manual, n.d.

[4] S. R, "Bipolar junction transistor and their applications," *Journal of Emerging Technologies and Innovative Research (JETIR)*, n.d.

[5] n.d., *Operational amplifier basics*, Unpublished technical manual, n.d.

[6] GeeksforGeeks. "Difference between diode and zener diode." Updated: Jul. 23, 2025. [Online]. Available: https://www.geeksforgeeks.org/difference-between-diode-and-zener-diode/

[7] I. Poole, *Understanding resistor specifications, parameters and specs*, Electronics Notes, Accessed: Jun. 2025, 2024. [Online]. Available: https://www.electronics-notes.com/articles/electronic_components/resistors/resistor-specifications-specs-parameters.php

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Mechatronics Engineering

Course Title	:	Electronics Sessional
Course Code	:	EEE 2188
Experiment No	:	02
Name of Experiment	:	Dui Nambar Exp
Date of Experiment	:	21 June 2026
Date of Submission	:	28 June 2026

Submitted By	**Submitted To**
Name : TASNIMUL HASAN Roll : 2408020 Series : 24	Md. Firoj Ali Associate Professor Department of Mechatronics Engineering RUET Prangon Das Assistant Professor Department of Mechatronics Engineering RUET

Experiment No: 2

Name of Experiment: Experiment 02

2.1 hi

hello

hello